Project: JWPCP Effluent Outfall Tunnel Project | Boring: **B-1A** | Pg. 1 of 7

Drilling Co.:	Cascade Drilling Inc.	Ground Elevation (Feet NAVD88):	36.6	Total Depth (Feet):	**131.0**
Drill Rig Type(s):	CME85	Location:	N 1749126.2	Date Started:	9/9/13
Drilling Method(s):	Hollow Stem Auger/ Rotary Wash		E 6475175.9	Date Completed:	9/11/13
Sampling Method:	Standard Split Spoon / California Modified	Depth to First Groundwater (Feet):	Not Measured ▽	Logged by:	J Martos
Borehole Diameter:	8 to 12.5'; 4.63 to 131'	Depth to Static Groundwater (Feet):	50.20 ▼	Reviewed by:	G. Miller, PE, CEG

Elevation (feet, NAVD88) (Surface) 36.60	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry Unit Weight (pcf)	Water Content (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
								Asphalt Concrete (5")									Hand auger to 5'
35		B-A						**ARTIFICIAL FILL (af) ?** Sandy Lean CLAY (CL) [0.42' - 3']: loose to medium dense, brown, moist, fine sand		15	60				SG R CR		
								LAKEWOOD FORMATION (Qlw) ? Sandy Lean CLAY to Lean CLAY with sand (CL) [3' - 10']: medium stiff to very stiff, dark grayish brown, moist, fine sand, some white veins									
	5	D-1		14/18	6 12 16	(18)		@ 5.5', stiff to very stiff, reduced sand content	114	15	78	39	26	4.5 P			
30		S-2		18/18	3 6 8	14		@ 7.5', clayey sand		19					CR	500+/2	
	10	D-3		18/18	4 8 14	(14)		Fat CLAY (CH) [10' - 21.5']: stiff to very stiff, olive brown mottled with brown and iron oxide staining, moist, some white pockets		28	96	66	43	4.5 P			
25		S-4		18/18	4 6 8	14											start mud rotary drilling at 12.5'
	15																

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
	05/31/17			04.62130050	B-2 - 1

Project: JWPCP Effluent Outfall Tunnel Project | **Boring:** B-1A | Pg. 2 of 7

Driller: Cascade Drilling Inc. | Drilling Method: Hollow Stem Auger/ Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
20		D-5		18/18	7 14 27	(26)		Fat CLAY (CH) [10' - 21.5']: stiff to very stiff, olive brown mottled with brown and iron oxide staining, moist, some white pockets (cont'd) @ 16', very stiff, pockets of pale brown		30	94	77	52	3.8 P			
		S-6		18/18	6 8 10	18										500+/2	
15	20	D-7		18/18	14 10 14	(15)		@ 20', olive brown with iron-oxide staining, lenses (<1"-thick) of pale brown clayey sandstone fragments						3.8 P			
								Silty SAND (SM) [21.5' - 35']: dense, olive brown, moist, fine to medium sand, with occasional thin lenses/layers of fine grained soils, with few shell fragments			14						
		S-8		12/18	7 14 22	36		@ 23', pale brown sand with silt								15/0.5	
10	25	D-9		18/18	16 28 35	(39)				11		NP	NP				
																	Pitcher barrel sample
	30																pressure meter test performed at 29'
5																	pressure meter test performed at 31'
	35	S-10		18/18	14 14 25	39		@ 32.5', stratified light brown, olive brown, with iron-oxide staining		17	25					150/1	

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 1

Project: JWPCP Effluent Outfall Tunnel Project | **Boring:** B-1A | Pg. 3 of 7

Driller: Cascade Drilling Inc. | Drilling Method: Hollow Stem Auger/ Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
0		D-11		18/18	20 27 35	(39)		Poorly graded SAND with clay (SP-SC) [35' - 37.5']: dense, olive brown, moist, fine to medium sand, some shells									
		S-12		18/18	10 10 18	28		Sandy Lean CLAY (CL) [37.5' - 40']: very stiff, olive brown, moist, micaceous, with lenses of gray clayey sand and iron-oxide staining @ 38.5', increased sand content, mottled with gray clayey sand		29	68	44	20			500+/0.5	
-5	40	D-13a D-13b		17.5/17.5	28 40 50/5.5"	(56/11.5")		Silty SAND (SM) [40' - 45.5']: very dense, mottled grayish brown and olive brown with iron-oxide staining, moist, fine sand, micaceous									
		S-14		18/18	18 18 27	45				26	20					54/1.5	
-10	45	D-15		11.5/11.5	28 50/5.5"	(31/5.5")		@ 45', with veins of brown lean clay Silty SAND (SM) to Poorly graded SAND with silt (SP-SM) [45.5' - 47.5']: very dense, light gray, moist, fine sand, micaceous									
		S-16		15/18	13 15 20	35		Silty SAND (SM) [47.5' - 52.5']: dense to very dense, mottled grayish brown and olive brown with iron-oxide staining, moist, fine, micaceous, lenses of brown lean clay		18	21					22/1.8	
-15	50	D-17		0/5.5	Ref/5.5"	(Ref/5.5")		@ 50', few shell fragments, olive brown below									no recovery with Cal-Mod, resampled with SPT and sand catcher at 50'
	55	S-18		16/18	21 30 36	66		Poorly graded SAND with silt (SP-SM) [52.5' - 67.5']: very dense, grayish brown, moist, fine sand, micaceous, weakly cemented								13/2.5	

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 1

Project: JWPCP Effluent Outfall Tunnel Project **Boring:** B-1A

Driller: Cascade Drilling Inc. Drilling Method: Hollow Stem Auger/ Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
								Poorly graded SAND with silt (SP-SM) [52.5' - 67.5']: very dense, grayish brown, moist, fine sand, micaceous, weakly cemented (cont'd)									
-20																	pressure meter test performed at 56.5'
																	pressure meter test performed at 58.5'
	60	S-19		17.5 / 17.5	30 39 50/5.5"	89/11.5"				20	12					135 / 7	
-25																	
		S-20		15 / 18	30 38 36	74		@ 62.5', lenses of medium sand									
	65	S-21		16 / 18	29 39 28	67		@ 66', seam of olive brown lean clay to 66.5'								150 / 5.5	
-30																	
		S-22		18 / 18	19 26 40	66		Silty SAND (SM) [67.5' - 81']: very dense, olive brown to grayish brown, moist, fine sand, micaceous		25					CR		
	70	S-23		10 / 18	20 29 41	70			100	26	16				DS	275 / 5.5	
-35																	
		S-24		11.5 / 11.5	30 50/5.5"	50/5.5"											
	75																

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 1

Project: JWPCP Effluent Outfall Tunnel Project
Boring: B-1A

Driller: Cascade Drilling Inc.
Drilling Method: Hollow Stem Auger/ Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
		S-25		16/18	40 40 46	86		Silty SAND (SM) [67.5' - 81']: very dense, olive brown to grayish brown, moist, fine sand, micaceous (cont'd)								350/4.5	
-40																	
		S-26		11.5/11.5	40 50/5.5"	50/5.5"				19	12						
	80																
								Sandy CLAY (CL) to Sandy SILT (ML) [81' - 92.5']: stiff to very stiff, mottled olive brown, grayish brown, and iron-oxide staining, moist, fine sand									
-45																	pressure meter test performed at 81.5'
																	pressure meter test performed at 83.5'
	85	P-27		15/15				@ 85.5', hard									
-50		D-28		18/18	10 20 24	(28)		@ 86.5', less silt, increased sand content	99	26	46	45	26	4.3 P			
		S-29		5/18	5 12 30	42		@ 88', lense of sand with silt, dense, light grayish brown, fine to medium sand, trace mica								500+/1.5	
	90	S-30		7/11.5	49 50/5.5"	50/5.5"											
-55																	
		S-31		18/18	12 19 42	61		Silty SAND (SM) [92.5' - 131']: very dense, light grayish brown with iron-oxide mottling, moist, fine sand, lenses of lean clay (>1.5" thick) @ 93.5', olive yellow		30	27					500+/2.5	
	95																

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Borehole Log for JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 1

Project: **JWPCP Effluent Outfall Tunnel Project** Boring: **B-1A**

Driler: Cascade Drilling Inc. Drilling Method: Hollow Stem Auger/ Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
		S-32		11.5/11.5	35, 50/5.5"	50/5.5"		Silty SAND (SM) [92.5' - 131']: very dense, light grayish brown with iron-oxide mottling, moist, fine sand, lenses of lean clay (>1.5" thick) (cont'd)									
								@ 95', decreased silt content, light grayish brown with iron-oxide mottling, micaceous									
-60																	
		S-33		17.5/17.5	29, 38, 50/5.5"	88/11.5"				28	15						
	100																
-65																	pressure meter test performed at 101.5'
																	pressure meter test performed at 103.5'
	105	S-34		11.5/11.5	28, 50/5.5"	50/5.5"				29	17					500+/11	
-70																	
		S-35		11/11	31, 50/5"	50/5"		@ 107.5', grayish brown									
	110	S-36		10.5/11.5	33, 50/5.5"	50/5.5"				27	16					290/7	
-75																	
		S-37		11.5/11.5	30, 50/5.5"	50/5.5"											

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 1

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Project: JWPCP Effluent Outfall Tunnel Project | **Boring:** B-1A |

Driller: Cascade Drilling Inc. | Drilling Method: Hollow Stem Auger/ Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
		S-38		16/18	26 28 30	58		Silty SAND (SM) [92.5' - 131']: very dense, light grayish brown with iron-oxide mottling, moist, fine sand, lenses of lean clay (>1.5" thick) (cont'd) @ 115', gray, increased silt content		27	23					4/11	
-80																	
		S-39		5.5/5.5	Ref/5.5"	Ref/5.5"		@ 117.5', decreased silt content									
	120	S-40		5.5/5.5	Ref/5.5"	Ref/5.5"										3.5/11	
-85																	
		S-41		5/5	Ref/5"	Ref/5"		@ 122.5', decreased silt content		20	9						
	125	S-42		0/11.5	39 50/5.5"	50/5.5"										2.8/5.5	
-90																	
		S-43		2/5.5	Ref/5.5"	Ref/5.5"		@ 127.5', with few shell fragments									
	130	S-44		11.5/11.5	47 50/5.5"	50/5.5"		@ 130', grades to silty sand		23	14					2/6	
								Boring drilled to a depth of 131 feet. 2" observation well installed on 9/11/13.									End drilling at 131' on 9/11/13.
-95																	

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 1

Project: **JWPCP Effluent Outfall Tunnel Project** | Boring: **B-1B** | Pg. 1 of 12

Drilling Co.:	Cascade Drilling Inc.	Ground Elevation (Feet NAVD88):	34.9	Total Depth (Feet):	**220.5**
Drill Rig Type(s):	CME85	Location:	N 1748778.5	Date Started:	8/26/13
Drilling Method(s):	Rotary Wash		E 6475214.2	Date Completed:	9/6/13
Sampling Method:	Standard Split Spoon / California Modified	Depth to First Groundwater (Feet):	Not Measured ∇	Logged by:	J Martos
Borehole Diameter:	4.63 to 220.5'	Depth to Static Groundwater (Feet):	48.20 ▼	Reviewed by:	G. Miller, PE, CEG

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry Unit Weight (pcf)	Water Content (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
(Surface) 34.90		B-A						**ARTIFICIAL FILL (af) ?** Sandy SILT (ML) [0' - 4.9']: stiff, reddish brown, moist		4	58	35	6				Hand auger to 5'
		G-1		6/6		Ref/6"											
30	5							@ 4.8', yellowish brown clayey sand									
								LAKEWOOD FORMATION (Qlw) ? Clayey SAND (SC) [4.9' - 13']: medium dense to dense, brown, damp, fine to medium sand									
		D-2		18/18	15 20 30	(31)											
		S-3		12/18	29 31 33	64		@ 7.5', yellowish brown, damp to moist, mostly fine sand below 7.5'								6.3/2.5	
25	10	D-4		10/18	14 18 20	(24)											
		S-5		18/18	8 15 21	36		Poorly-graded SAND with clay (SP-SC) [13' - 15']: dense, yellowish brown, moist, fine sand								7.5/2.1	
	15																

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
	05/31/17			04.62130050	B-2 - 2

Project: JWPCP Effluent Outfall Tunnel Project **Boring:** **B-1B** Pg. 2 of 12

Driller: Cascade Drilling Inc. Drilling Method: Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
		D-6		16/18	9 15 22	(23)		Sandy Lean CLAY to Lean CLAY (CL) [15' - 25']: very stiff, olive, moist, with small, black, hardened inclusions	108	22	86	44	27				
		P-7		36/36				@ 19', sandy lenses and pockets of iron-oxide staining									
15	20																
																	pressure meter test performed at 23.5'
10	25																
		D-8		18/18	6 10 15	(16)		Fat CLAY (CH) [25' - 31']: very stiff, moist, laminae of dark orange iron-oxide staining and concretions (small sandstone fragments)		36	96	85	50				
		S-9		18/18	6 9 12	21										500+/1.5	
5	30	P-10		18/18													
								Silty SAND (SM) [31' - 35']: dense, olive, wet, medium to coarse sand, few fine gravel									
		D-11		18/18	20 25 27	(33)		@ 32', grades to medium sand, weak calcium carbonate cementation		18	16	NP	NP				
		S-12		16/18	17 19 10	29		@ 33', with coarse sand								45/2.5	
	35																

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 2

Project: JWPCP Effluent Outfall Tunnel Project **Boring:** B-1B Pg. 3 of 12

Driller: Cascade Drilling Inc. Drilling Method: Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
		D-13		18/18	9 15 20	(22)		Fat CLAY with sand (CH) [35' - 38']: very stiff, olive yellow mottled with olive, moist, some silt	93	30	79	50	27		SG DS		
		S-14		18/18	13 16 18	34		Silty, Clayey SAND (SC-SM) [38' - 58']: dense to very dense, gray mottled with iron oxide staining, moist, fine sand								27/2.8	
-5	40							@ 40', becomes very dense, laminated sand and lean clay, grayish brown									
		D-15		17.5/17.5	20 21 50/5.5"	(44/11.5")			109	13					SG	6.5/1.5	
		S-16		16/18	12 18 34	52		@ 42.5', olive brown, clayey with trace fine gravel, few shell fragments to 45.5'			23	22	7				
-10	45	D-17		0/11.5	39 50/5.5"	(31/5.5")											D-17: no recovery, retrieved sample with SPT and sand catcher at 45'
-15	50																pressure meter test performed at 51'
		S-18		15/18	30 40 45	85				25	16					80/6.5	
	55																

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 2

Project: JWPCP Effluent Outfall Tunnel Project **Boring:** **B-1B** Pg. 4 of 12

Driller: Cascade Drilling Inc. Drilling Method: Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
								Silty, Clayey SAND (SC-SM) [38' - 58']: dense to very dense, gray mottled with iron oxide staining, moist, fine sand (cont'd)									pressure meter test performed at 56'
		S-19		10/11.5	33 50/5.5"	50/5.5"		Poorly graded SAND with silt (SP-SM) [58' - 68']: very dense, grayish brown, moist, fine to medium sand								165/3.5	
-25	60	S-20		16/18	37 24 27	51		@ 60', trace clay		22	11						
		S-21		18/18	27 28 22	50		@ 63.4', approximately 1"-thick lens of iron-oxide staining underlain by very fine sand with clay @ 63.5', clay in sampler shoe								2.5/6	
-30	65	D-22		18/18	37 38 45	(52)			101	24					SG DS		
		S-23		18/18	8 16 27	43		@ 67.5', 6"-thick layer of sandy lean clay, very stiff to hard, mottled gray and brown with iron-oxide staining Clayey SAND (SC) [68' - 71.5']: very dense, mottled gray and olive brown with iron oxide staining, moist, fine to medium sand		23	54	38	20			4.5/6	
-35	70	S-24		0/11.5	37 50/5.5"	50/5.5"											S-24: no recovery, resampled with SPT to retrieve sample
								Poorly graded SAND with silt (SP) [71.5' - 77.5']: very dense, grayish brown, moist, fine to medium sand									
		S-25		17.5/17.5	27 38 50/5.5"	88/11.5"		@ 72.5', dark gray seams, small pockets of iron-oxide staining								11/4	
	75																

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Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 2

Project: JWPCP Effluent Outfall Tunnel Project **Boring:** B-1B Pg. 5 of 12

Driller: Cascade Drilling Inc. Drilling Method: Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
		S-26		17.5 / 17.5	30 34 50/5.5"	84/11.5"		Poorly graded SAND with silt (SP) [71.5' - 77.5']: very dense, grayish brown, moist, fine to medium sand (cont'd) @ 75', olive yellow, medium grained sand, subangular to angular, some shell fragments in shoe		12	7						
		S-27		11.5 / 11.5	34 50/5.5"	50/5.5"		@ 77.5', with few shell fragments		15	7						
-45	80	S-28		11.5 / 11.5	36 50/5.5"	50/5.5"									AVS	11 / 5.5	
		S-29		11.5 / 11.5	36 50/5.5"	50/5.5"				18	7						
-50	85	S-30		15.5 / 18	26 36 40	76		@ 86', with some shell fragments below 86'								12 / 7	
-55	90																pressure meter test performed at 89' pressure meter test performed at 91'
		S-31		17 / 18	35 42 42	84		Silty SAND (SM) [92.5' - 177']: very dense, dark greenish gray, moist, fine sand, iron-oxide streaking		27	14					33 / 65	
	95																

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Geotechnical and Environmental Consultants

Borehole Log
for
JWPCP Effluent Outfall Tunnel Project

Approved	Date	Revised	Date	Project No.	FIG.
JK	05/31/17			04.62130050	B-2 - 2

Project: JWPCP Effluent Outfall Tunnel Project | **Boring:** B-1B | Pg. 6 of 12

Driller: Cascade Drilling Inc. | Drilling Method: Rotary Wash

Elevation (feet, NAVD88)	Depth (feet)	Sample No.	Sample Type	Recovered / Penetration	Blow Count (per 6")	SPT N-Value / Core RQD	Graphic Log	Material Description	Dry UW (pcf)	Water Cont. (%)	% Passing #200	Liquid Limit (%)	Plasticity Index	Undrained Shear Strength (ksf)	Other Tests	PID/FID (ppm)	Remarks
		S-32		17.5/17.5	29 40 50/5.5"	90/11.5"		Silty SAND (SM) [92.5' - 177']: very dense, dark greenish gray, moist, fine sand, iron-oxide streaking (cont'd)		30	17						
		S-33		17.5/17.5	28 40 50/5.5"	90/11.5"		@ 98.5', grades to poorly graded sand with silt								22/18	
-65	100	S-34		11.5/11.5	36 50/5.5"	50/5.5"		@ 100', increased silt content		25	22						
		S-35		15.5/17.5	30 40 50/5.5"	90/11.5"		@ 103', decreased silt content								3.5/14	
-70	105	S-36		11/11.5	38 50/5.5"	50/5.5"		@ 105', no visible iron-oxide streaks or veins below									
		S-37		11.5/11.5	36 50/5.5"	50/5.5"				26	18					13/6.5	
-75	110	S-38		11.5/11.5	30 50/5.5"	50/5.5"											
		S-39		5.5/5.5	Ref/5.5"	Ref/5.5"										88/6	

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